Code No: 136FU **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year II Semester Examinations, July - 2023
REMOTE SENSING AND GIS
(Common to EEE, ME, ECE, CSE, IT)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

PART – A

(25 Marks)

1.a)	Write down fundamentals of stereoscopy.	[2]
b)	What is parallax in photogrammetry?	[3]
c)	Define active remote sensing.	[2]
d)	What is spatial resolution?	[3]
e)	Write the purpose of datum.	[2]
f)	What do you mean by projection?	[3]
g)	Write topology rules.	[2]
h)	Define raster.	[3]
i)	What is metadata?	[2]
j)	What is remote sensing data?	[3]

PART – B

(50 Marks)

2.a) Explain principles of photogrammetry.
b) List and explain various types of aerial photograph along with the neat sketches. [5+5]

OR

3. What is relief displacement and how it is useful in photogrammetry and how it is measured. [10]

4. Elaborate the following:
a) Spectral resolution
b) Radiometric resolution
c) Temporal resolution. [10]

OR

5.a) Draw the electromagnetic spectrum along with band widths and its utility.
b) Compare natural colour composite and false colour composite. [5+5]

6. Explain the following:
a) Datum
b) Map projections
c) Coordinate systems. [10]

OR

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7.a) Write down the components of GIS.
b) Differentiate spatial data and attribute data. [5+5]

8.a) Write the importance of down topology rules.
b) Compare raster data and vector data. [5+5]

OR

9.a) Explain object based vector data model.
b) Explain classes and their relationship. [5+5]

10.a) Write down elements of the raster data model.
b) Explain types of raster data and raster data structures. [5+5]

OR

11.a) Elaborate integration of raster and vector data.
b) Explain how field data is integrated with remote sensing data. [5+5]

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